

General revision

II — ratio, percentage and rates

Sharing, proportion, the three percentage questions, percentage change and speed.

LESSON 195–198

4 × 40 MIN

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S7 WHOLE-COURSE REVIEW

LESSON CLOCK

30'

35'

35'

35'

20'

5

Ratio	30 min
Sharing and proportion	35 min
The three percentage questions	35 min
Percentage change	35 min
Speed and motion	20 min
Homework	5 min

LEARNING OBJECTIVES

- Simplify a ratio and share a quantity in a given ratio.
- Use direct and inverse proportion correctly.
- Answer all three kinds of percentage question, including the reverse one.
- Use $s = vt$ and find an average speed.

TEXTBOOK

National	pp. 557–568
Cambridge	Stage 7, pp. 120–132
Workbook	Checkpoint practice 2

01

Explanation

Ratio

RATIO	DIVIDE BY	SIMPLEST FORM
12 : 18	6	2 : 3
45 : 60	15	3 : 4
0.5 : 2	multiply by 2 first	1 : 4
$\frac{3}{4} : \frac{1}{2}$	multiply by 4 first	3 : 2

A RATIO COMPARES PARTS; A FRACTION COMPARES A PART WITH THE WHOLE

In $2 : 3$ the first quantity is $\frac{2}{5}$ of the total, not $\frac{2}{3}$. Adding the parts before writing any fraction removes the commonest error of the whole topic.

Sharing and proportion

TASK	PARTS	ONE PART	ANSWER
share 60 in $2 : 3$	5	12	24 and 36
share 84 in $3 : 4$	7	12	36 and 48
share 120 in $2 : 3 : 5$	10	12	24, 36 and 60

KIND	SITUATION	METHOD	ANSWER
direct	5 pens cost 20 000 so'm; 8 pens?	one pen is 4000	32 000 so'm
inverse	4 workers take 12 days; 6 workers?	the job is 48 worker-days	8 days

MORE WORKERS, FEWER DAYS

In direct proportion both quantities rise together; in inverse proportion one rises as the other falls, and the *product* stays the same. Ask which of the two the situation is before calculating.

The three percentage questions

QUESTION	EXAMPLE	METHOD	ANSWER
find a percentage of a number	15% of 240	$0.15 \cdot 240$	36
express one number as a percentage of another	30 out of 120	$\frac{30}{120} \cdot 100$	25%
find the whole from a percentage	20% of it is 45	$45 \div 0.2$	225

NAME THE QUESTION BEFORE CHOOSING THE METHOD

All three use the same three numbers in different places. Writing “this is the third kind” at the top of the working is worth more than remembering three formulas.

Percentage change

$$\text{multiplier} = 1 + \frac{\text{percentage}}{100} \text{ for a rise, } 1 - \frac{\text{percentage}}{100} \text{ for a fall}$$

TASK	MULTIPLIER	WORKING	ANSWER
increase 400 by 15%	1.15	$400 \cdot 1.15$	460
decrease 500 by 20%	0.8	$500 \cdot 0.8$	400
200 rises to 250	—	$\frac{50}{200} \cdot 100$	a 25% rise
80 falls to 60	—	$\frac{20}{80} \cdot 100$	a 25% fall
after a 20% rise a price is 60 000	1.2	$60\,000 \div 1.2$	50 000 so'm

THE CHANGE IS ALWAYS MEASURED AGAINST THE ORIGINAL

200 to 250 is a rise of 25%, but 250 back to 200 is a fall of 20%. The same difference, two different percentages, because the starting numbers differ.

Speed and motion

$$s = vt \quad v = \frac{s}{t} \quad t = \frac{s}{v}$$

QUESTION	WORKING	ANSWER
180 km in 3 h	$180 \div 3$	60 km/h
4 km in 20 min	$4 \div \frac{1}{3}$	12 km/h
7 km at 5 km/h	1.4 h	1 h 24 min
72 km/h in m/s	$72 \div 3.6$	20 m/s
120 km at 60, back at 40	$240 \div 5$	48 km/h
two cars 300 km apart at 60 and 40	$300 \div 100$	meet after 3 h

UNITS FIRST, FORMULA SECOND

Minutes become hours before they meet a speed in km/h; an average speed is total distance over total time and never the mean of the speeds.

02 Worked examples

EXAMPLE 1 Share 120 in the ratio 2 : 3 : 5.

① $2 + 3 + 5 = 10$ parts.

② $120 \div 10 = 12$ for one part.

③ 24, 36 and 60.
They add to 120 ✓

ANSWER

24, 36 and 60

EXAMPLE 2 After a 20% rise a price is 60 000 so'm. Find the original price.

① The multiplier for a 20% rise is 1.2.

② The original times 1.2 is 60 000.

③ $60\,000 \div 1.2 = 50\,000$ so'm.
Check: $50\,000 \cdot 1.2 = 60\,000$ ✓

ANSWER

50 000 so'm

EXAMPLE 3 A car drives 120 km at 60 km/h and returns at 40. Find the average speed.

- ① Out 2 h, back 3 h.
- ② Total 240 km in 5 h.
- ③ 48 km/h.
Not 50 — more time at the slower speed.

ANSWER

48 km/h

03 Interactive model

Put the three percentage questions on the board with the same three numbers in each and let the class sort them; naming the question is the skill being revised.

Ratio, percentage and rates in eight questions

12 : 18 in simplest form:

2 : 3

3 : 2

6 : 9

4 : 6

Sharing 60 in 2 : 3 gives:

20 and 40

24 and 36

30 and 30

12 and 18

15% of 240 is:

16

36

24

3.6

30 as a percentage of 120 is:

4%

25%

30%

40%

Increasing 400 by 15% gives:

415

460

340

600

After a 20% rise a price is 60 000; it was:

48 000

50 000

54 000

72 000

4 km in 20 minutes is:

0.2

12

80

4

4 workers take 12 days; 6 workers take:

18

8

6

12

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Ratio	Nisbat	Отношение
To share in a ratio	Nisbatda taqsimlash	Делить в отношении
Direct proportion	To'g'ri proporsionallik	Прямая пропорциональность
Inverse proportion	Teskari proporsionallik	Обратная пропорциональность
Percentage	Foiz	Процент
Percentage change	Foizli o'zgarish	Процентное изменение
Multiplier	Ko'paytuvchi	Множитель
Average speed	O'rtacha tezlik	Средняя скорость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In the ratio 2 : 3 the first part is:

$\frac{2}{3}$ of the total

$\frac{2}{5}$ of the total

half

$\frac{3}{5}$ of the total

More workers taking fewer days is:

direct proportion

inverse proportion

a ratio

a percentage

The multiplier for a 20% fall is:

0.2

0.8

1.2

20

A percentage change is measured against:

the new value

the original value

the difference

a hundred

Average speed is:

the mean of the speeds

total distance over total time

the largest speed

the first speed

72 km/h in m/s is:

7.2

20

26

259

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Simplify $12 : 18$

ANSWER $2 : 3$

2. Simplify $45 : 60$

ANSWER $3 : 4$

3. Share 60 in the ratio $2 : 3$

ANSWER 24 and 36

4. 15% of 240

ANSWER 36

5. 180 km in 3 hours

ANSWER 60 km/h

6. 30 as a percentage of 120

ANSWER 25%

7. 5 pens cost 20 000 so'm: one pen

ANSWER 4000 so'm

Medium · the standard question

7 PROBLEMS

1. Share 84 in the ratio 3 : 4

ANSWER 36 and 48

2. Share 120 in the ratio 2 : 3 : 5

ANSWER 24, 36 and 60

3. 8 pens at 4000 so'm each

ANSWER 32 000 so'm

4. Increase 400 by 15%

ANSWER 460

5. Decrease 500 by 20%

ANSWER 400

6. 4 km in 20 minutes

ANSWER 12 km/h

7. 72 km/h in m/s

ANSWER 20 m/s

Hard · stretch and reason

7 PROBLEMS

1. Simplify $\frac{3}{4} : \frac{1}{2}$

ANSWER 3 : 2

2. 4 workers take 12 days: 6 workers

ANSWER 8 days

3. 200 rising to 250

ANSWER A 25% increase

4. 80 falling to 60

ANSWER A 25% decrease

5. After a 20% rise a price is 60 000 so'm: the original

ANSWER 50 000 so'm

6. 120 km at 60 km/h and back at 40

ANSWER 48 km/h

7. Two cars 300 km apart at 60 and 40 km/h meet after

ANSWER 3 hours

07 Homework

Homework — 5 tasks

Name the kind of question at the top of each solution before working it.

- Share 150 in the ratio 3 : 7 and 180 in the ratio 1 : 2 : 3.
- Find 18% of 350, and express 42 as a percentage of 120.
- A price falls by 15% to 68 000 so'm. Find the original price.
- 6 workers take 10 days. How long would 4 workers take?

5. A cyclist rides 30 km at 15 km/h and 30 km at 10. Find the average speed.